

Using Smart Glasses to Support Situated Decision Making in Aquaculture

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ABSTRACT

The rapidly expanding prawn farming industry is vital in providing high-quality protein for the increasing global population. The key to consistently high yield is the efficient and effective management of pond water, maintaining the optimal growth environment. Our consultations with seven Australian commercial prawn farms found that the lack of access to essential real-time water information and trends has hindered farmers from making confident and timely decisions while in the field, which contributes considerably to either prolonged suboptimal pond conditions or increases in operational costs. This paper describes two aquaculture pond water management tasks that can benefit from wearable technology and discusses general concerns about using smart eyewear in harsh outdoor farming environments. A *Pondside Visualisation System* was created to allow farmers to access visualised historical and real-time water data during field operations using Google Glass Enterprise Edition 2. The application uses a new data visualisation style that is optimised for small near-eye displays. A validation study was conducted with seven industrial practitioners (including a supervisor and six farm technicians) on a commercial farm with real data from the recent growing season, where each technician made 100 pond management decisions using our system. Objective measurements of task completion time and task accuracy indicated the farmers achieved an accuracy of 86.4% and 89.2% for two management tasks with at least 41.0% less time compared to existing desktop-based practice. A structured expert review confirmed the usability of our system and discussed methods to mitigate issues discovered during the study. We also discussed lessons learned from the project.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; **Field studies**; Empirical studies in visualization; • **Applied computing** → **Agriculture**.

KEYWORDS

smart glasses, aquaculture, decision-support, smart farming, in-situ analysis, empirical study

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1 INTRODUCTION

Aquaculture is critical in providing high-quality protein to the booming population. According to the Food and Agriculture Organization of the United Nations (FAO), global aquaculture production (food products only) reached 80 million tons in 2016. Taking the prawn farming industry as an example, the annual production value is estimated to be AU\$66 million in Australia [15] and US\$531 million in the US [16]. A small improvement to this industry is a big contribution to our society.

Our consultations with seven major Australian commercial fisheries discovered two critical daily farm operational tasks that could be enhanced by smart glasses. The first one is the frequent pond water quality sampling via hand-held sensing devices. The second one is the follow-up analysis of the collected water quality data. Both tasks are vital to successful pond management and are generally performed under great time pressure. This paper is interested in the latter one, where smart glasses can potentially improve the efficiency and effectiveness of the pond analysis task by providing contextualised pond water quality information.

Smart glasses have made remarkable differences in logistics [11, 12, 14] education [10, 19], manufacturing [17, 21], health and medicine [10, 19, 22]. However, there is an extremely limited number of research studies exploring the potential of smart glasses in farming activities. Previous studies include using smart eyewear for accessing

cattle and milk production information on a cattle farm [20], presenting farm information sheet on a dairy farm [5, 6], navigating farmers when performing soil sampling jobs [13], and assisting plant chlorophyll measurement activities [7]. Although limited in quantity, these studies suggest that smart glasses, with the heads-up, hands-free way of retrieving and processing information in real-time, could contribute to enhanced task performance.

This paper presents the first empirical study that introduces smart glasses to the aquaculture farming industry. We are particularly interested in enabling pond-side farm management decision-making by providing farmers with essential water quality information via smart glasses. The paper discusses the challenges in selecting smart eyewear and visualising farm data on small near-eye displays, as well as proposing practical guidelines for them. A wearable application system, that enables aquaculture farmers to access and analyse the essential water quality information in the field, was described and trialled with industrial practitioners on a commercial prawn farm. We also discuss the lessons we learned from working with the aquaculture industry. The findings in this study should also apply to all pond-based aquaculture farms as water analysis task is a common practice as suggested by FAO [1, 3].

2 CONTEXT AND MOTIVATION: POND WATER ANALYSIS TASK

Based on consultations with seven commercial farms, involving site visits, interviews and focus groups, we learned that pond data analysis has to be performed manually by highly-skilled farmers via examining the visualisations of water quality variables. As reported by both the farms and FAO, the two most important pond analysis tasks are:

Task 1 Determine if the aeration in a pond needs to be adjusted [1].

Task 2 Determine if the water in a pond needs to be exchanged [1].

This is often referred to as “flow the pond”.

Both tasks require the farmers to access historical water data, which is also referred to as “pond context”. However, farmers can only access such pond context using a desktop computer in the office after manually sampling the entire farm. As a result, farmers have to travel back to the field to adjust aeration and water flow, causing delayed reactions that directly result in unnecessary operational expenses, such as labour and electricity. For example, it costs the farm over A\$700¹ per hour to just run aerators on a 100-hectare facility. In addition, pond environments are fast changing with dynamics that are extremely hard to predict. For example, in certain extremely rare situations, the dissolved oxygen level can drop from 6.2 mg/L (normal) to 1.5 mg/L (catastrophic) in just two hours. Thus, a timely response must be made to avoid any loss of stock. These negative impacts could have been avoided/minimised if the farmers had access to the pond context at the pondside during the sampling task. This naturally leads to a smart glasses-based solution that has a unique advantage in enabling hands-free access to relevant task context [12]. This removes the delay between pond sampling and analysis, as the analysis task can be performed parallelly to the sampling task.

¹This is calculated based on 4 aerators (10 horsepower each) per hectare, with an electricity rate of 25 cent/kWh.

3 PROPOSED POND SIDE VISUALISATION SYSTEM

We propose the *Pondside Visualisation System* that allows farmers to access the pond context and perform both the *Task 1* (i.e., aeration adjustment) and *Task 2* (i.e., water inlets adjustments) right near the pond in a hands-free way.

While the examples and experiments presented in this paper used data from Farm P, the system is compatible with pond data from all aquaculture farms. There was no conflict of interest involved when choosing which farm to develop and trial our *Pondside Visualisation System* with. We chose Farm P because it is a large leading commercial farm, and more importantly, it has established good practices of maintaining historical water quality data. The rest of this section describes the challenges, design, implementation and deployment of the *Pondside Visualisation System*.

3.1 System Design and Implementation

3.1.1 Conventional Pond Data Analysis. Water data analysis is mostly done by examining line plots generated in spreadsheet software, such as Excel. An actual Excel-based visualisation tool used by Farm P is shown in Figure 1. The multi-variable line plot is used by the farmers when making pond management decisions, such as aeration and water exchange control. Each variable, instead of being organised as a single time series, is split into two series, i.e., the AM and PM reading. This indicates that each pond is sampled twice a day, with one in the morning (approx. 4 am) and one in the afternoon (approx. 2 pm). Each data point is labelled with its value as farmers need to check the readings (values) against certain thresholds when assessing the possibility of a pond crash².



Figure 1: A screenshot of the pond data visualisations used by a commercial farm (farm branding elements were occluded).

A System Usability Survey (SUS) [2] survey suggested that farmers have polarised attitudes toward the usability of the Excel-based system. Apart from T3, both T1 (32.5) and T2 (35.0) gave relatively low scores for the current Excel-based tool (see Appendix A). On-site interviews with farmers reported a series of usability issues with the conventional spreadsheet tool, including 1) label size was too small to read; 2) translating raw data from the hand-held unit into the spreadsheet was “terrible” due to intensive copying and pasting; 3) switching between ponds was an “annoying”.

3.1.2 Design of Water Quality Visualisation. To the best of the authors’ knowledge, there is no current guide on visualising data on smart glasses. We found that interactively visualising data on smart glasses is extremely challenging due to the ultra-small display

²A pond crash refers to a sudden drop in one or more pond variables, such as a dissolved oxygen crash.

and the lack of precise navigation control. This is because most interactions on smart glasses are done via physical buttons/touchpads and/or voice input, which requires substantial effort from the wearer to simply move the cursor to a specific point on the screen [23].

We proposed a practical guideline on presenting static line plots in an ultra-small smart glasses display, based on several user-centred design iterations with direct inputs from prawn farmers. The guidelines include:

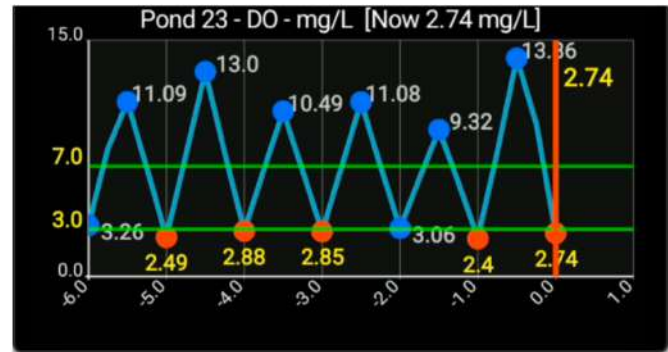
- **Relative Timeline:** Instead of using the absolute timestamp, a wearable visualisation system should use a relative timeline, which is sufficient for presenting water quality trends. The use of a relative timeline was also supported by the farm manager at Farm P in a pilot study.
- **Single variable plot:** Multi-variable plots appear to be heavily cluttered on the smart glasses display, thus should be split into multiple single-variable plots.
- **Shortened Reporting Period:** A long reporting period makes the chart too cluttered to be readable. An informal user study was conducted to evaluate the readability of the graph against the length of the reporting period on a Google Glass. The study indicated that the line plot can hold up to 14 vertices with label text of 23sp³ in bold style. Based on the study, we reduced the reporting period down to seven days, i.e., six-day historical data and one-day forecast data (requires additional forecast model, e.g., [8, 9]).
- **Critical Threshold Indicator:** Multiple horizontal lines that mark the optimal range of a given water variable help prawn farmers identify concerning readings instantly.
- **Coloured Annotations:** The data point and its label should be highlighted if its value is alarming.

Based on the design guideline, we de-cluttered the original plot generated by Excel on a PC (see Section 3.1.1) and added marker lines to highlight the data points of interest. Working with farmers from Farm P, numerous visualisation styles were generated. The final version is presented in Figure 2. Other selected prototypes are listed in Appendix B.

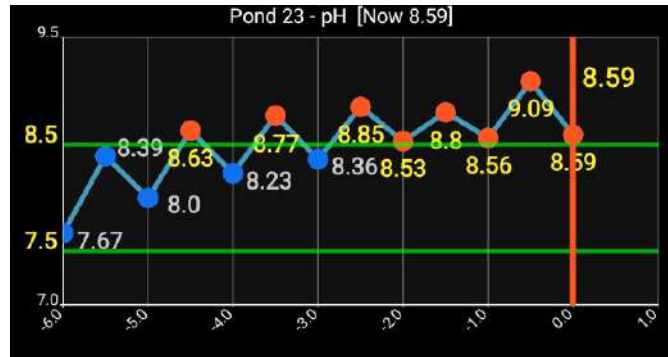
3.1.3 System Implementation. The *Pondside Visualisation System* was implemented using Java and Kotlin in Android Studio. A pond decision logging component was also implemented to log the farmers' pond management decisions. This system is also used in the validation study presented in Section 4. Figure 3 shows a wireflow of the system.

The application starts with a welcome screen. Pond selection activities will be triggered by tapping the touchpad or a voice command. A user can swipe the touchpad and tap to select which pond to check. Then the system will retrieve the historical data from Senaps (a cloud-based sensory data management platform) and show a graph on the screen. After checking the visualised water variable, the user can swipe the touchpad to view the decision logging screen, where the user will be asked two questions, i.e. "Leave the aeration on?" (i.e., *Task 1*) and "Flow the pond?" (i.e., *Task 2*). The default answers to both questions were set to "No". Finally, a confirmation screen with the user's decisions will be shown before the user proceeds to the next pond.

³A scalable pixel (sp) is a unit for specifying the size of a font in the Android system.



(a) Six-day historical dissolved oxygen data is plotted.



(b) Six-day historical pH data is plotted.

Figure 2: Example visualisations of water quality variables, with highlighted alarming data points.

The visualisation component was implemented using Android Plot (an open-source plotting package). As the package does not support customising the paint attributes (e.g., colour and size) of a single point in a series, multiple layers (based on the order of drawing method calls) have to be used to draw the graph, such background layer, normal/abnormal data layers and normal/abnormal data annotation layers. The touchpad gestures (e.g., tap and swipe) were implemented using native Android touchpad APIs.

3.2 System Deployment

The *Pondside Visualisation System* has to be deployed to smart eyewear that can survive the harsh aquaculture farming environment. However, there were no well-defined guidelines on selecting appropriate smart eyewear suitable for aquaculture tasks or the driving factors behind the guidelines. To understand the practical challenges, we interviewed six farmers (four at Farm P and two at Farm B) and shadowed four farmers (three at Farm P and one at Farm B, approximately 2 hours with each farmer) when they were performing routine pond management tasks. Two main concerns raised by the farm technicians were usability and durability.

3.2.1 Practical Challenges on Aquaculture Farms. For durability, aquaculture facilities are typically built in the coastal area, where the high humidity and chlorine-ion in the salt-water environment

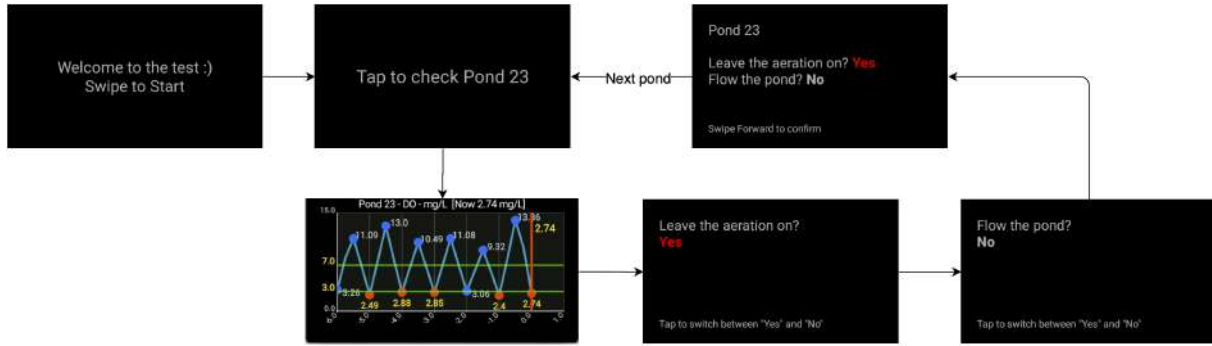


Figure 3: The wireflow for the *Pond Visualisation System*.

can cause rapid moisture corrosion and salt encrustation and damage the smart eyewear.

For general usability, our interviews reported that disruption, discomfort, privacy, and health and safety are the top four concerns when using smart glasses on an aquaculture farm. **Disruption** refers to any negative impacts that may be caused by the use of a device (e.g., extended task completion time in water sampling and analysis activities). **Discomfort** refers to any physical stress caused by prolonged usage wearing a heavy device (e.g., eye fatigue and ear muscle strain). Aquaculture farms have unique **Safety** concerns, such as accidentally dropping a device in the pond when performing sampling tasks on narrow piers.

Although these factors are not detailed here, researchers and engineers should also consider other concerns, such as hygiene issues when using a shared device and the variety of accessories, such as external camera and storage, prescription and polarised lens, charging speed, and connectivity. General wearability attributes should also be considered, such as placement of the device on a human body, thermal and accessibility [18].

3.2.2 A New Smart Eyewear Selection Criteria. We propose the following criteria when selecting smart eyewear for aquaculture tasks.

- **Ruggedisation:** This is essential against the harsh saltwater environment that causes moisture corrosion and salt encrustation.
- **Weight:** The device should be lightweight to minimise any discomfort. The weight should be as close as possible to that of a pair of normal glasses.
- **Computing power:** Visualising large datasets is computing-intensive due to intensive data parsing and canvas drawing. High computing power reduces latencies between user inputs and system responses (i.e., drawing visualisations), which is a major usability indicator. It also enables the capability of running machine learning/deep learning packages (e.g., TensorFlow Lite), which is a trending demand in the smart farming era.
- **Camera:** The camera is not only required for keeping visual pond records (e.g., photo of prawn), but essential for running advanced computer vision algorithms (normally requires

certain image resolution). This is another trending demand in smart farming.

- **Operating system and developer support:** This directly impacts the design, development, deployment and maintenance of the software applications.

3.2.3 Google Glass Enterprise Edition 2. Based on the criteria in Section 3.2.2, we evaluated Google Glass XE-C and Enterprise Edition 2 (Glass EE2), Dream Glass, and Epson BT-300. Glass EE2 was chosen for this project, as it is water and dust resistant (IP53 rated), untethered, and lightweight (46g, close to a pair of sunglasses). It also comes with a faster CPU and a decent camera with 8 megapixels and 80 degrees of field of view. It also offers various accessories, such as a safety/prescription frame that can be fitted with sunglasses lens, which is highly desired in aquaculture facilities. System-wise, Glass EE2 runs on Android Oreo (AOSP 8.1.0), whereas others mostly run on Android 4.4.1 (KitKat). The newer version is not only more developer-friendly, but also provides a longer battery life and fast USB-C charging support. It is also suitable for long-term wearing in the mobility condition, which was confirmed by a technician, who did not report any discomfort when asked after wearing the Glass EE2 for 3 hours.

4 ON-FARM VALIDATION METHODOLOGY

This section describes the protocol used to validate the effectiveness and efficiency of the *Pondside Visualisation System*. Only dissolved oxygen levels, the dominating factor for the morning pond analysis task, were used in the study. The study was approved by the CSIRO Social Science Human Ethics Research Committee (Approval No. 105/18).

4.1 Research Questions

The validation study focuses on the two following research questions:

- Q1** Does the *Pondside Visualisation System* allow prawn farmers to make pond management decisions as **accurately** as the current best practice?
- Q2** Does the *Pondside Visualisation System* allow prawn farmers to spend the **same or less time** in making the same number of pond management decisions than the current best practice?

4.2 Participants and Procedure

Six farm technicians from the Farm P and their manager participated in the study. Prawn farmers are a small and highly specialised expert group and the number of participants was based on local standards [4]. The study was set up and conducted remotely due to the COVID-19 restrictions. A video conference briefing was given to the participants at the beginning of the study. The main study consists of two parts:

Part 1: requires each farm technician ($N = 6$) to perform the two tasks described in Section 2, i.e., determine if the aerators (*Task 1*) and water inlets (*Task 2*) need to be adjusted using the *Pondside Visualisation System* (see Figure 4a and 4b). All farm technicians have at least 3 months of experience in using handheld devices to collect water quality data, as well as analysing the data using their current spreadsheet tool. Each farmer will examine 50 ponds and will not move from the initial pond. The water quality data used in the study was collected by Farm P two months prior to the study, which eliminates any potential learning impact.

Part 2 requires the farm manager to examine the same 50 ponds (DO only) and make the same decisions but only using the spreadsheet tool. The farm manager's reference answers serve as the ground truth for *Q1*.

4.2.1 Data Collection. The following objective measures are considered to answer the two questions:

- Q1** Data collected includes decisions made by farmers using the *Pondside Visualisation System* (Part 1) and the farm manager using the existing spreadsheet tool (Part 2).
- Q2** Data collected includes system-logged Task Completion Time (TCT) during Part 1.

For subjective measures, we conducted a structured expert review with farmers to further understand other benefits and impacts that are difficult to quantify.

5 RESULTS

Six farm technicians, identified as T1 to T6, completed their system evaluation sessions. However, T5 experienced a technical issue that all decisions were recorded under the same pond, making the data not usable. As a result, this section only discusses the data collected from the other five technicians.

5.1 Task Accuracy

The accuracy for each task is calculated by comparing the decisions made by each technician against the reference answers made by the manager. Overall, the mean accuracies of both tasks are highly impressive with 89.2% (*Task 1*) and 86.4% (*Task 2*) respectively. **This indicates that the farmers were able to make pond management decisions using our systems as accurately as the existing tool (Q1).**

With each technician examining 50 ponds, the average accuracy of *Task 1* is 89.2% or 44.6 of 50 ponds (see Table 1). On average, 26 of 50 ponds needed aeration. The result demonstrates that the technicians were able to make the same decisions as they were using the spreadsheet tool on a desktop computer nearly 90.0% of the time. Also, the variance among the users is considerably low with a standard deviation of only 2.0% or 1 out of 50 ponds.

The second task is determining if the water in a pond needs to be exchanged. On average, 8 of 50 ponds needed water change. According to Table 2, the average accuracy is 86.4% or 43.2 of 50 ponds. The accuracy across five technicians is also highly consistent with a low deviation of approximately 3.2% or 1.6 of 50 ponds.

We also checked the number of default answers made by the farmers. On average, the aeration task had 52% (130/250) of the decisions being the default, while the water exchange task had 85.2% (213/250). The accuracy of the aeration task (with fewer default answers) showed a higher accuracy, which suggests no obvious relation between the number of default answers and the task accuracy.

5.2 Task Completion Time

Task completion time (TCT) measures the total time a user spent examining 50 ponds (see Table 3). To assess the efficiency of the *Pondside Visualisation System*, the TCT of using the *Pondside Visualisation System* were compared with the average time of using the spreadsheet tool. The latter one is approximately 12 minutes and 30 seconds as reported by the farm manager during the interview (see Section 2).

According to Table 3, the mean time spent examining 50 ponds and making 100 decisions is 7 minutes 20 seconds using the *Pondside Visualisation System*. When compared to that of using the spreadsheet tool (12 minutes 30 seconds for 50 ponds), using our system is approximately 41.3% (5 minutes 10 seconds) faster. **The result indicates that prawn farmers spent less time making the same pond management decisions using our *Pondside Visualisation System* than using the spreadsheet tool (Q2).**

5.3 Post-Study Interview and Survey

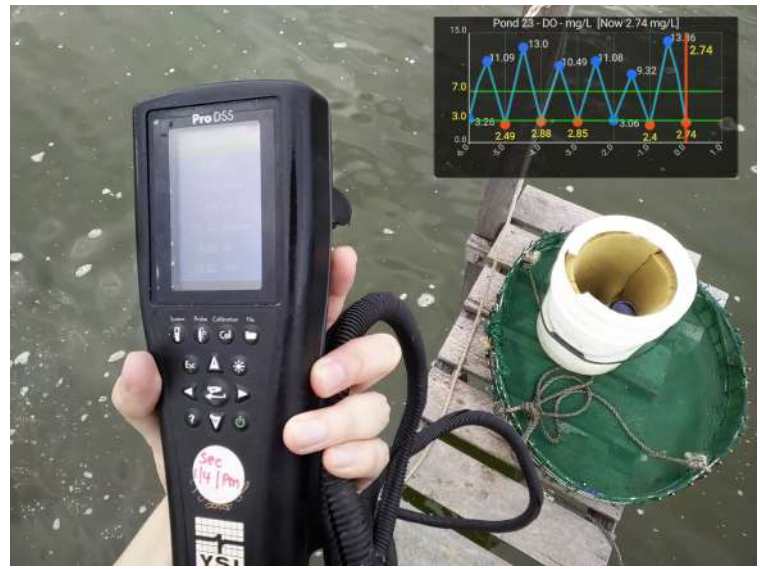
Post-study interviews with both the manager and the technicians were conducted to further understand the results of the study and collect feedback for future system improvement. The interview focused on understanding the cause of differences in decision accuracies and how that may impact the farm operation.

5.3.1 Interview with the Farm Manager. We interviewed the manager with four questions listed in Appendix C. According to his responses, the technicians also make different decisions with the existing spreadsheet tool. This is because farm decisions are highly dynamic in nature. One example he noted was that if a day is considerably busy, the technicians have to cherry-pick the most alarming ponds and drop several less concerning ones, as the farm did not have enough workforce to handle all of them. Most importantly, all the alarming water quality issues were correctly identified and responded to (i.e., correct decisions were made) in the study. The manager believed that the main cause of the minor inconsistency was the lack of training, as such inconsistency was also observed when the technicians were firstly trained to use the spreadsheet tool, which was eventually resolved with more training over time.

One obvious benefit confirmed by the manager is that the technicians can make more informed decisions in a timely manner. He also confirmed that being able to respond to pond conditions early can surely reduce operational costs, such as electricity and help the business to better manage the workforce.



(a) A technician is using the *Pondside Visualisation System* near a production prawn pond.



(b) The first person view of using the *Pondside Visualisation System*. The Google Glass visualisation appears near the top right corner of the wearer's view.

Figure 4: Using the *Pondside Visualisation System* near a pond.

Table 1: Task Accuracy – Task 1: Aeration Adjustment Decision ($N = 50$).

	T1	T2	T3	T4	T6	Mean	SD
Count of Correct Ponds	43	45	46	44	45	44.6	1.0
Accuracy	88.0%	86.0%	92.0%	90.0%	90.0%	89.2%	2.0%

Table 2: Task Accuracy – Task 2: Water Exchange Control Decision ($N = 50$).

	T1	T2	T3	T4	T6	Mean	SD
Count of Correct Ponds	45	41	43	45	42	43.2	1.6
Accuracy	90.0%	90.0%	86.0%	82.0%	84.0%	86.4%	3.2%

Table 3: Task completion time (24 HR).

	T1	T2	T3	T4	T6	Mean
Start Time	16:07:03	10:56:33	15:26:25	14:58:55	12:14:16	-
End Time	16:16:51	11:05:08	15:34:31	15:05:42	12:17:39	-
Duration	0:09:48	0:08:35	0:08:06	0:06:47	0:03:23	0:07:20

5.3.2 Interview with Farm Technicians. Six technicians were interviewed with questions listed in Appendix D. The interview reported a mixed experience with the *Pondside Visualisation System*. Most technicians thought it was “cool” to use wearable tech on the farm and agreed that the system could help them make more informed decisions during sampling runs. However, the voice recognition feature was reported to be frustrating as the device struggled to recognise the keywords when the operator was facing a strong wind or has a strong accent. One technician noted that “*The system*

was able to pick up the voice command most of the time. But we had to try many times when we were facing the wind.” In contrast, navigating the system using the touchpad on the Glass was described as “easy-to-use” and “just works”.

One technician asked us to add pH visualisations to the system so she can start it as a production system for daily pond analysis tasks. Another technician reported a fitting issue where he could not see the entire screen of the Glass. He had to spend extra effort to read the plots, which made his eyes uncomfortable. This is an issue

that will be resolved with more training and proper instructions. No other discomfort was reported.

No specific concerns were initially reported by the technicians. Then we extended the question to ask if they have privacy concerns as the Glass has recording capabilities. The technicians reported no concerns about seeing co-workers wearing the Glass in the facility. Using the camera or microphone for work activities, such as analysing prawn weight using computer vision or making a voice note, was also not a concern. However, the technicians are concerned if the Glass streams live audio or video to the management.

6 DISCUSSION

6.1 Lessons Learned

Simple visualisations are required on a smart glass to support decision-making tasks when designed following our guidelines. As we discussed in Section 3.1.2, interactive visualisation is fundamentally difficult to achieve on wearable eyewear due to limited interaction support. We designed our system to only present static graphs, which were validated by expert users to be effective. Researchers and engineers can use our guidelines as a starting point when designing their applications in the future.

Alternatives to voice input, such as touchpads or hand gestures, are required in an outdoor farm environment. Our experience suggests that an on-farm test of the input component (e.g., voice), in an interactive system, must be tested before the main system is evaluated. We did numerous tests on the voice input component in both a lab environment with 70 dB wind noise (recorded at the farm) played through a speaker and an urban environment with busy traffic. The speech engine responded correctly to nearly all attempts while there was only one speaker (the wearer). However, the voice input still struggled as reported in the post-study survey and was eventually abandoned by all participants.

On-farm studies need to be highly flexible compared to tightly-structured lab studies, where the study environment and the participants' engagement can normally be well controlled. Different levels of interruptions were experienced in over fifteen formal and informal studies we conducted with farms. For example, an unexpected storm hitting the coastal area will automatically postpone a study, as all farmers will be busy fixing the alarming ponds. This reflects a natural gap in designing on-farm user experiments. Researchers normally have less knowledge and control over the farm environment, while farmers have minimum knowledge of how a scientific user evaluation works. We suggest that an on-farm study needs to be interruption-resistant with loosely coupled components and a non-linear procedure.

6.2 Limitations

Although every effort has been made by the authors to ensure the on-farm validation study is both scientifically sound and practically feasible, there are limitations to the study. Although our system is demonstrated to be effective and efficient, the participants were all expert users. It is unclear if a novice user (e.g., a trainee) can achieve the same level of performance without additional support.

6.3 Future Work

The system and validation study described in this paper point to a range of potential further investigations. These include the use of additional outdoor tracking mechanisms, such as GPS or Bluetooth Low Energy (BLE) Beacons, that will allow the system to automatically localise which pond the operator is examining. Customised voice recognition should be explored to consider whether it can effectively reduce the impact of wind noise. There is scope to consider whether more feature-rich head-mounted displays, such as Microsoft HoloLens 2 can bring additional practical benefits. There is an opportunity to quantify the suitability and the number of hours of training that might be required to further explore and eliminate usability issues and stabilise the decision-making variance in this type of system. And deeper understanding is needed around how to most efficiently interact with graphs displayed on smart glasses.

7 CONCLUSION

The *Pondside Visualisation System* presented in this article is an innovative way to provide aquaculture farmers with essential pond conditions and trends required to make informed decisions in the field. The system can be effortlessly tweaked to fit other aquaculture crops (e.g., fish, crab and oyster) by supplying relevant water data and domain knowledge (e.g., critical thresholds for each water condition). The measurements of task accuracy and task completion time have strongly validated that the system is both effective and efficient in supporting farmers with their pond analysis tasks. Additionally, the expert reviews suggested that adopting wearable technology like our system can lead to the reduction of operational expenses.

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A SYSTEM USABILITY SURVEY

The results of SUS are shown in Table 4.

Table 4: SUS results from the three participants (ordered by the SUS score).

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Score
T3	5	3	5	2	4	1	3	2	5	2	80.0
T2	3	4	2	2	2	4	2	4	3	4	35.0
T1	2	4	2	2	1	4	2	5	4	3	32.5

B SELECTED VISUALISATION DESIGNS

Four selected line plot variations are presented in Figure 5.

C QUESTIONS TO THE MANAGER

Questions asked to the manager are:

- (1) On several occasions, your technicians made different decisions for the same pond. For example, three technicians decided to turn on the water valves for Pond No. 30, 86 and 92, while two technicians and you decided not to do so. Why do you think that happened?
- (2) Are those different decisions detrimental to your business?
- (3) What would you do to mitigate this?
- (4) What are the potential benefits do you think the wearable technology can bring?

D QUESTIONS TO THE TECHNICIANS

Questions asked to the technicians are:

- (1) How was your overall experience using the system?
- (2) What are the things you like and dislike about the visualisation?
- (3) What new features would you like to see in the future?
- (4) Did you experience any discomfort during your session?
- (5) Do you have any concerns over wearing a smart glass at the workplace?

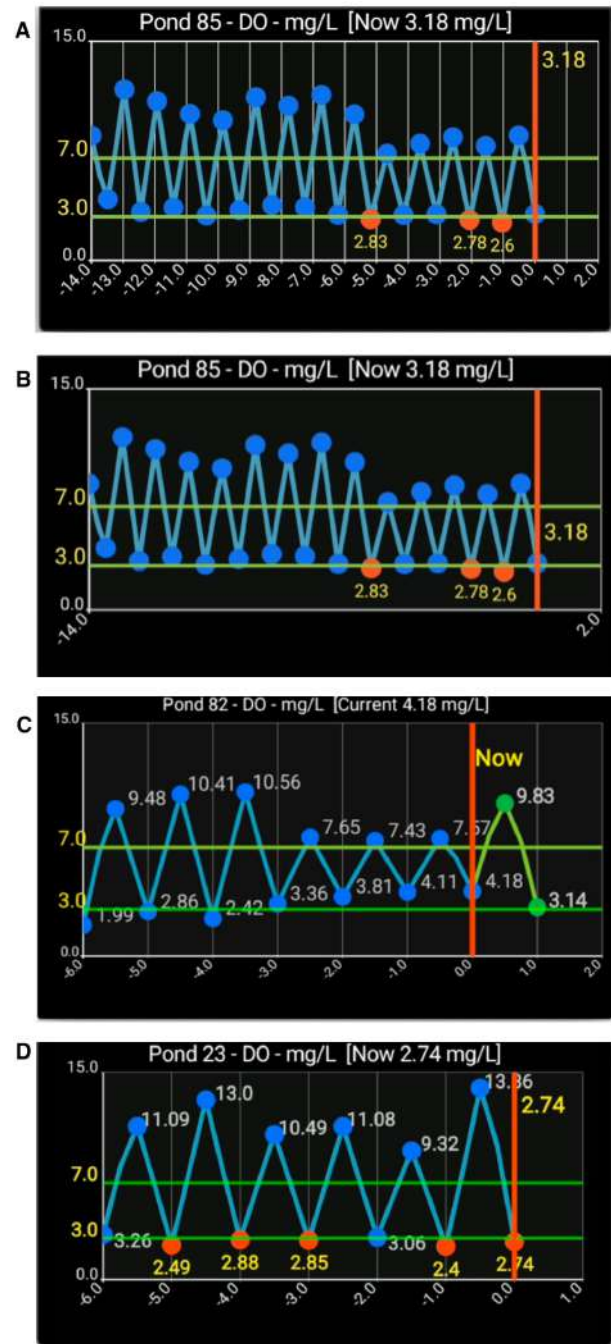


Figure 5: (A) Fourteen-day historical data. (B) Fourteen-day historical data with only alarming data points highlighted. No grids and x-axis labels. (C) Six-day historical and two-day forecast data with labels. (D) Six-day historical data with labels. Alarming data points are highlighted.